

Actuarial Internet-Interruption Risk

Business loss, network resilience and conditional provisions in Tunisia

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1 Executive summary

This study develops a disclosure-safe actuarial framework for self-reported economic loss associated with internet interruptions among formal businesses represented by Tunisia's 2024 World Bank Enterprise Survey (WBES). It combines survey-weighted estimation, a conditional Gamma duration model, repeated cross-validation, a governed machine-learning challenger, public telecommunications evidence and explicit 2027-2031 planning scenarios.

The usable connected-business sample contains 616 establishments and 136 reported disruption events. The survey-weighted disruption rate is 19.8%. Among disrupted businesses, the loss-ratio field has 117 usable responses and only 16 positive values. This small severity base is the dominant limitation.

The calibrated expected loss ratio is 0.252% of annual-sales exposure. Its weighted-resampling interval is 0.093% to 0.458%. For a standardized 10 million TND exposure, the persistent-condition

scenario gives a 2027 risk-adjusted provision of 30 293 TND. The 2031 systemic-stress illustration reaches 77 273 TND.

These amounts are conditional management risk budgets. They are not national loss estimates, insurance premiums, statutory technical provisions, booked reserves or forecasts of any telecommunications operator.

2 Scope, data and estimand

2.1 Restricted business survey

The source file is the Tunisia WBES 2013-2020-2024 panel, but the internet-interruption module used here is available only for the 2024 wave (World Bank Enterprise Surveys 2025). Raw microdata remain local. Only aggregate results satisfying disclosure controls are published.

The 2024 module supplies:

- disruption occurrence;
- average disruption hours and minutes;
- losses as a percentage of annual sales;
- monetary losses;
- difficulty switching providers;
- annual fixed-broadband cost;
- survey weight and establishment characteristics.

The primary occurrence estimand is the weighted share of connected formal establishments with a valid response that reported an internet disruption in the previous fiscal year. It is not a network uptime statistic and does not cover households or informal businesses.

2.2 Public network context

Two official reports from the Instance Nationale des Telecommunications (INT) are collected with URLs and SHA-256 checksums. The 2019 mobile campaign measured 3G/4G service in Grand Tunis for Tunisie Telecom, Ooredoo Tunisia and Orange Tunisia (Instance Nationale des Telecommunications 2019). The 2020 fixed campaign evaluated 8 Mbps ADSL service across 24 governorates and five internet service providers (Instance Nationale des Telecommunications 2021).

A current public RIPE Atlas probe snapshot describes the available measurement footprint by autonomous system (RIPE Network Coordination Centre 2026). Probe counts are not market share, service quality or outage frequency.

2.3 Separation rule

The sources are deliberately kept separate:

WBES business loss $\nRightarrow$ operator attribution,

because WBES does not identify the provider. INT and RIPE Atlas evidence provides system context but is never joined to individual business records.

3 Descriptive evidence

3.1 Occurrence, duration and switching

Table 1: Core 2024 survey evidence

Measure	Estimate
Raw establishment records	645
Valid connected-business responses	616
Effective occurrence sample	251.6
Reported disruptions	136
Weighted disruption rate	19.8%
Usable duration responses	129
Weighted mean duration	2.0 hours
Weighted provider-switch constraint rate	21.5%

Hours and minutes are combined as

$$D_i = \begin{cases} H_i, & H_i > 0, \\ M_i/60, & H_i = 0 \text{ and } M_i \text{ observed,} \\ \text{missing,} & \text{otherwise.} \end{cases}$$

This rule preserves the survey's less-than-one-hour structure and does not turn unknown minutes into zero.

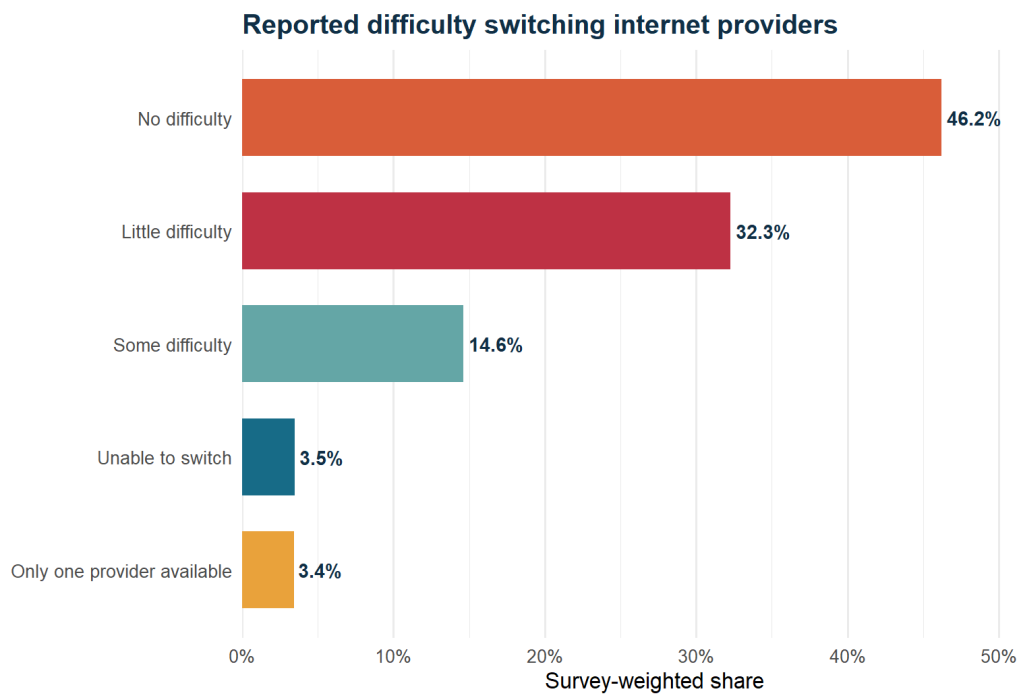


Figure 1: Survey-weighted provider-switching responses

The switching indicator groups one-provider-only, unable-to-switch and some-difficulty responses. It is a descriptive resilience indicator, not a causal predictor or competition-law conclusion.

3.2 Loss-cost components

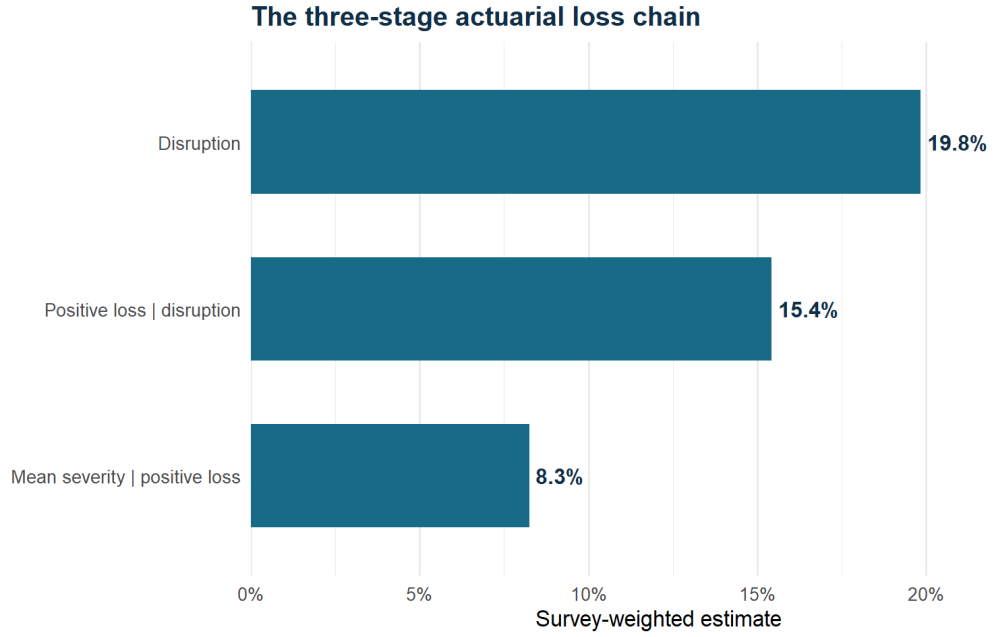


Figure 2: Three-stage actuarial loss chain

Table 2: Survey-weighted actuarial loss-chain calibration

Component	Estimate
Disruption probability	19.8%
Positive loss probability given disruption	15.4%
Mean positive loss ratio	8.259%
Expected loss ratio	0.252%

4 Actuarial framework

4.1 Survey-weighted occurrence

For establishment i , let $I_i \in \{0, 1\}$ be the disruption indicator and w_i its survey weight. Following standard design-based estimation (Lumley 2010),

$$\hat{p} = \frac{\sum_{i=1}^n w_i I_i}{\sum_{i=1}^n w_i}, \quad n_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

The occurrence GLM is

$$I_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = \beta_0 + \beta_R^\top R_i + \beta_S^\top S_i + \beta_Z^\top Z_i,$$

with region, sector and firm size as categorical covariates. The specification is intentionally parsimonious relative to 136 events.

4.2 Conditional duration

For $D_i > 0$, duration is capped at its 99th percentile:

$$D_i^* = \min(D_i, Q_{0.99}), \quad D_i^* \mid I_i = 1 \sim \text{Gamma}(\mu_i, \phi), \quad \log \mu_i = x_i^\top \gamma.$$

The cap reduces leverage from the maximum 48-hour response while retaining the record. Duration is conditional on having a reported interruption and a usable positive duration.

4.3 Expected loss ratio

The loss-cost decomposition follows the frequency-severity logic used in non-life actuarial work (Ohlsson and Johansson 2010):

$$\widehat{ELR}_{2024} = \hat{p} \hat{q} \hat{\mu}_+,$$

where

$$\hat{p} = \widehat{\Pr}_w(I = 1), \quad \hat{q} = \widehat{\Pr}_w(L > 0 \mid I = 1), \quad \hat{\mu}_+ = \widehat{\mathbb{E}}_w[L \mid L > 0, I = 1].$$

Numerically,

$$0.198268 \times 0.154168 \times 0.082587 = 0.0025244.$$

A monetary-severity model is rejected because only eight monetary values exist. A percentage-loss severity GLM is also rejected because only 16 strictly positive responses exist.

4.4 Weighted resampling uncertainty

For $b = 1, \dots, 2000$, a pseudo-population sample is drawn with replacement using

$$\pi_i = \frac{w_i}{\sum_j w_j}.$$

The complete three-stage estimator is recomputed in each sample. The percentile interval is

$$\left[Q_{0.025}\{\widehat{ELR}^{(b)}\}, Q_{0.975}\{\widehat{ELR}^{(b)}\} \right] = [0.0009333, 0.0045780].$$

This interval represents sampling variation under the resampling design. It excludes reporting error, model-form uncertainty and scenario uncertainty.

5 Validation and AI governance

5.1 Repeated cross-validation

Because the internet module is available only in 2024, no genuine temporal holdout is possible. Both occurrence models use repeated stratified five-fold cross-validation. Probability error is measured with the weighted Brier score (Brier 1950):

$$BS_w = \frac{\sum_{i \in \mathcal{V}} w_i (I_i - \hat{p}_i)^2}{\sum_{i \in \mathcal{V}} w_i}.$$

Weighted AUC measures discrimination. A training-fold weighted prevalence provides the Brier benchmark.

Table 3: Repeated-cross-validation model comparison

Metric	Actuarial GLM	Gradient boosting
Weighted Brier	0.1508	0.1614
Prevalence Brier	0.1600	0.1597
Weighted AUC	0.6924	0.6617
Observed rate	0.1993	0.1986
Predicted rate	0.1992	0.1990

Gradient boosting is a legitimate non-linear challenger for actuarial loss modelling (Guelman 2012). Here it uses region, sector, firm size, log annual sales and log broadband cost. Provider-switching answers are excluded to avoid using a potentially outcome-related response as a predictive shortcut.

5.2 Evidence gates

Table 4: Predeclared model and attribution gates

Component	Evidence	Minimum	Decision
occurrence_glm	136	100	Pass
duration_gamma	125	100	Pass
severity_glm	16	50	Block
ml_challenger	136	200	Block
operator_loss_attribution	0	1	Block

The AI challenger remains evaluation-only because the event count is below 200. Its permutation importance is cross-validated and diagnostic; it does not identify causal drivers.

6 Public network evidence

6.1 INT fixed benchmark

Table 5: INT 2020 fixed-internet national means for 8 Mbps ADSL

Metric	Value	Assessment
Download throughput	6.59 Mbps	Good
Upload throughput	0.83 Mbps	Good
Network latency	41.31 ms	Within threshold
DNS resolution time	62.8 ms	Within threshold
Access availability	98.62 %	Within 98% threshold
VoIP packet-loss rate	0.023 ratio	Outside threshold

The report found a 98.62% mean access-availability rate and described it as meeting the 98% threshold. The same report identified the VoIP packet-loss measure as outside its threshold. These results are specific to the 2020 ADSL campaign and cannot validate the 2024 WBES outcome.

6.2 INT mobile scope

Table 6: INT 2019 mobile QoS campaign measurement volume

Operator	Measurement attempts	Geography	Technology
Tunisie Telecom	3723	Grand Tunis	3G/4G
Ooredoo Tunisia	4011	Grand Tunis	3G/4G
Orange Tunisia	3667	Grand Tunis	3G/4G

These counts document campaign scope. They are not service rankings and are not combined with loss responses.

7 Conditional projection and provision

Let $E_{s,2027}$ be starting annual-sales exposure, h_s exposure growth, m_s initial risk stress, g_s annual risk trend and λ_s prudence margin. Then

$$E_{s,t} = E_{s,2027}(1 + h_s)^{t-2027},$$

$$\widehat{ELR}_{s,t} = \widehat{ELR}_{2024} m_s (1 + g_s)^{t-2027},$$

$$P_{s,t} = E_{s,t} \widehat{ELR}_{s,t} (1 + \lambda_s), \quad C_{s,T} = \sum_{t=2027}^T P_{s,t}.$$

Table 7: Projection assumptions

Scenario	Initial exposure	Exposure growth	Initial stress	Risk trend	Prudence
Resilience improvement	10 000 000 TND	2.0%	0.9x	-3.0%	10.0%
Persistent conditions	10 000 000 TND	2.0%	1x	0.0%	20.0%
Systemic stress	10 000 000 TND	2.0%	1.6x	6.0%	40.0%

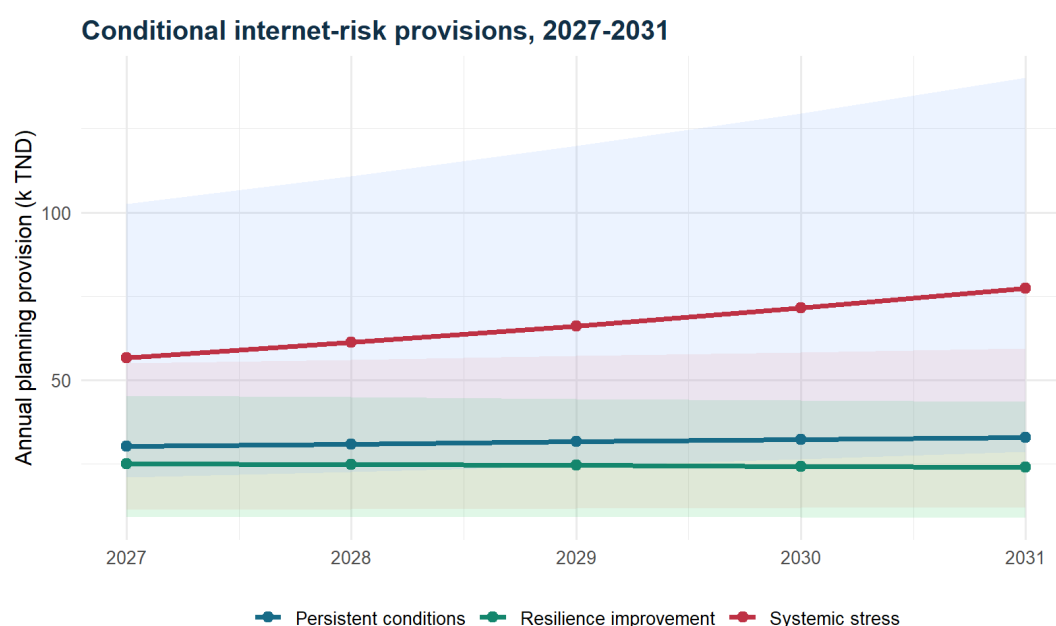


Figure 3: Conditional annual provision paths

Table 8: 2031 annual and cumulative provision per 10 million TND starting exposure

Scenario	ELR	Annual provision	Lower	Upper	Cumulative
Resilience improvement	0.201%	23 949 TND	8 854 TND	43 431 TND	122 336 TND
Persistent conditions	0.252%	32 790 TND	12 123 TND	59 465 TND	157 645 TND
Systemic stress	0.510%	77 273 TND	28 569 TND	140 137 TND	332 530 TND

The shaded interval propagates only the empirical 2024 resampling interval. Scenario uncertainty is represented through separate paths rather than mixed into a single probabilistic claim.

8 Limitations and interpretation

The analysis has five material limitations.

First, the outcome is self-reported and recalled over a fiscal year. Second, the internet module is available only for 2024, preventing temporal validation. Third, severity is sparse and heavy-tail estimates are unstable. Fourth, no WBES provider field exists, so operator-level loss attribution is impossible. Fifth, public network context is dated and heterogeneous: a 2019 Grand Tunis mobile campaign, a 2020 national ADSL campaign and a current volunteer-probe snapshot do not measure the same population or event.

The most defensible interpretation is therefore portfolio conditional: given a defined annual-sales exposure and explicit stress assumptions, the calibrated loss ratio provides a transparent starting point for continuity planning. It does not establish legal responsibility or future operator performance.

9 Reproducibility and controls

The targets pipeline orchestrates restricted-data harmonisation, survey evidence, modelling, validation, bootstrap calibration, projections and public artifacts. `renv` locks the R environment. Python scripts collect checksummed public context and evaluate the non-linear challenger. Unit tests verify missing-code handling, duration construction, weighted AUC and actuarial identities. A release check enforces segment thresholds and fails if identifiers, individual annual sales or individual monetary losses enter public CSV files.

10 Conclusion

The 2024 survey indicates a material internet-continuity signal among formal businesses, but the small positive-loss sample rules out precise severity modelling. The project responds with disciplined model governance: use a parsimonious occurrence GLM, model conditional duration, quantify empirical loss-cost uncertainty, evaluate but do not promote an unsupported AI challenger, keep operator context separate, and make all future assumptions visible.

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